



A novel Markov chain method for predicting granular mixing process in rotary drums under different rotation speeds

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ABSTRACT

The granular mixing in rotary drums has been intensively studied using discrete element method (DEM). However, DEM is computationally intensive. When the operating condition (such as rotation speed) of the drum is changed, a new round of time-consuming DEM simulation must be performed, whereas the data generated in previous rounds are not utilized. In order to solve this problem, we propose a Hermite interpolation-based Markov chain method (HI-MCM) to replace DEM simulation, so that granular mixing process under new drum speeds can be quickly predicted. HI-MCM utilizes the data generated by only several DEM simulations to establish Markov chain models, with which the granular mixing process at new rotation speeds can be then quickly predicted by Hermite interpolated Markov chain models. Tests demonstrate that the particle spatial distribution and mixing degree predicted by HI-MCM agrees well with DEM simulation results, while the computing time is less than 1% of that needed by DEM and current Markov chain methods. Moreover, the proposed HI-MCM shows good robustness to the change of Markov chain parameter (learning time step). The proposed method has potential usages in online prediction of granular mixing in rotary drums.

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1. Introduction

Rotary drums are widely used in metallurgical and chemical industry for thermal treatment of granular materials [1,2]. The mixing process of the granular material in the drum greatly affects the product quality and energy consumption, and has been therefore intensively studied in the last three years [3,4].

Some researchers have used different experimental techniques to investigate granular mixing in rotary drums, such as positron emission particle tracking [5], dyed-tracer [6,7], image analysis [8,9], radioactive particle tracking [10], nuclear magnetic resonance [11]. However, these techniques have limitations in particle numbers being tracked, or radiation risk, high cost, etc. In contrast, Discrete Element Method (DEM) is very powerful in simulating granular mixing on particle level [12–14]. However, DEM simulation is computationally intensive because the trajectories of all particles have to be calculated. According to literature report, the ratio of simulation time to physical time is about 1332:1 for a system with 10,875 particles [15], 1200:1 for a system with 32,000 particles [16]. Quasi real-time simulation (9.4:1) was reported in literature [17] for a large granular system with 9.6 million particles,

however, 270 high performance GPUs were needed for the simulation, which is rather expensive. Another shortage of DEM simulation is that, when the rotation speed of the drum is changed, such computation-intensive DEM simulations have to be performed again to obtain new results, whereas the data generated in previous rounds of simulation are not utilized.

In order to improve computing efficiency, some research proposed Markov chain models [18–21] as supplementary method to predict granular mixing process. Doucet et al. [22] proposed a DEM-based multidimensional state-space Markov chain model (DEM-MCM), using data from previous DEM simulations. This model was successfully used in predicting the mixing of 48,000 particles in a rotary drum at fixed rotation speed. On a computer cluster (including 8 Intel Xeon 3.6 GHz CPUs), DEM-MCM costs only 1 to 10 min to simulate 48 s of physical time, while DEM needs about 1 h of computing time. Tjakra, et al. [23] performed several DEM simulations for different rotation speeds and then constructed corresponding Markov chain operators. The Markov chain model was then able to predict mixing process at varying drum speeds by switching among the operators. However, the values of drum speed in the model should be the same as that already adopted in previous DEM simulations, otherwise, time-consuming DEM simulation has to be performed again to construct the corresponding Markov chain operator. In fact, rotation speed is an operating variable that is often adjusted in practice to control granular mixing process in rotary

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drums. To our best knowledge, current Markov chain models are only valid for predicting granular mixing under specified drum speeds. When drum speed is changed, the established Markov chain models will fail.

In order to solve this problem, we will propose a Hermite interpolation-based Markov chain method (HI-MCM) for fast prediction of granular mixing process under new drum speeds. HI-MCM utilizes the known dataset (data generated by only several DEM simulations) to establish Markov chain models, with which the mixing process (in terms of particle spatial distribution and mixing degree) at new rotation speeds can be then quickly predicted by Hermite interpolated Markov chain models, without performing additional DEM simulations.

2. Data generation by DEM

2.1. DEM simulation

In order to describe our proposed method, the mixing process of a mono-sized granular system is used as example in the present work to generate particle trajectory data, as shown in Fig. 1. The system contains 2674 red particles and 2674 white ones with the same size (particle radius: 10.3 mm), which are put side-by-side in a drum (radius: 606.4 mm; length: 89 mm) at initial state to observe the mixing process. To eliminate the end-wall effect, both ends of the drum are set as periodic boundaries [24]. This configuration also helps generate enough data for modelling by using small granular system [25]. Rotation speeds of the drum are configured between 4 rpm and 16 rpm so that the particle motion is in the rolling mode [26]. Other simulation parameters listed in Table 1 are adopted from our previous study [27] and will be therefore not repeated here.

2.2. Known dataset

With the parameters in Table 1, we performed 6 sets of DEM simulations (No.1–No.6 in Table 2) on a PC computer (Intel® Xeon® Processor E5–2640 v2 and 32GB RAM) with PFC^{3D} software. In each simulation, the drum speed is fixed ($n = 4, 6, 8, 10, 14, 16$ rpm) and

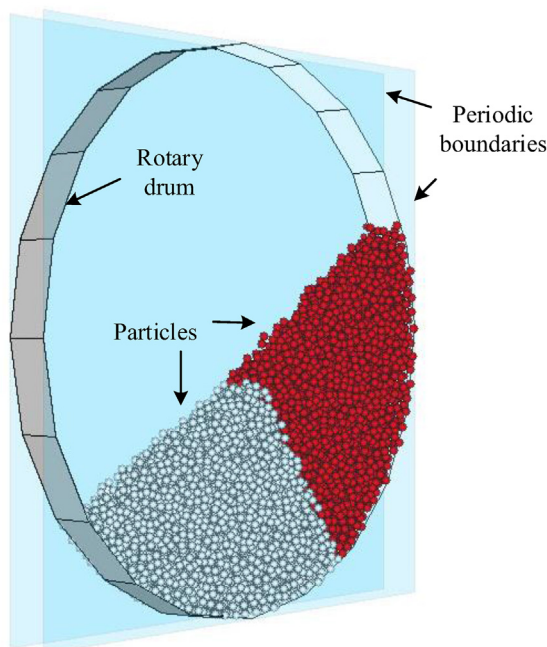


Fig. 1. Initial state of the granular system and the periodic boundaries.

Table 1
Parameters used in DEM simulations.

	Parameters	Value	Unit
Drum	Radius	606.4	mm
	Length	89	mm
	Rotation speed n	4–16	rpm
	Filling ratio	39.3	%
	Friction coefficient	0.28	–
Particle	Radius	10.3	mm
	Particle number	2674 red + 2674 white	–
	Density	1414.7	kg·m ⁻³
	Shear modulus	0.85	GPa
	Poisson's ratio	0.35	–
	Friction coefficient	0.56	–
	Critical damping ratio	0.15	–
	Contact model	Hertz–Mindlin	–

Table 2
DEM simulations and the computing time needed.

No.	Rotation speed n (rpm)	Simulated physical time t_1 (s)	DEM computing time t_2 (min)	Time ratio t_2/t_1 (–)
1	4	600	4262.5	426
2	6	400	3107.7	466
3	8	300	2528.8	506
4	10	240	2116.3	529
5	14	172	1595.6	557
6	16	150	1395.9	558

the simulation stops when the granular mixing reaches well-mixed state (about 40 drum revolutions). The particle trajectories at different time were saved in computer, with a sampling interval of 0.2 s.

It can be seen from Table 2 that the ratio of computing time to physical time is rather large (from 426:1 to 558:1) for the small granular system investigated in our work. If the system is larger, the required computing time will drastically increase. Moreover, when the drum speed is changed, a new round of time-consuming simulation has to be performed again, while the data from previous DEM simulations are not used.

In order to solve the above problems, we use results of the six DEM simulations as known dataset to analyze the particle movements, and then propose a Hermite interpolation-based Markov chain method (HI-MCM). With the established HI-MCM model, the mixing process at new rotation speeds can be quickly predicted, without performing additional time-consuming DEM simulations.

3. The proposed method

The schematic diagram of the proposed method (HI-MCM) is shown in Fig. 2. First, the Markov chain models for the known dataset should be established. The steps are as follows: a) construct the Markov chain operator by analyzing the simulated particle trajectories with cell number N and learning time step t_i ; b) calibrate the Markov time step by quantitatively comparing the mixing curve from a temporary Markov chain model with that from DEM simulation; c) establish the Markov chain model corresponding to each drum speed in the known dataset by performing step a) and step b). Second, establish the new Markov chain model for the new rotation speed by Hermite interpolation, with which the mixing curve as well as the particle spatial distribution can be predicted. The pseudo code of the proposed algorithms can be found in the Appendix A.

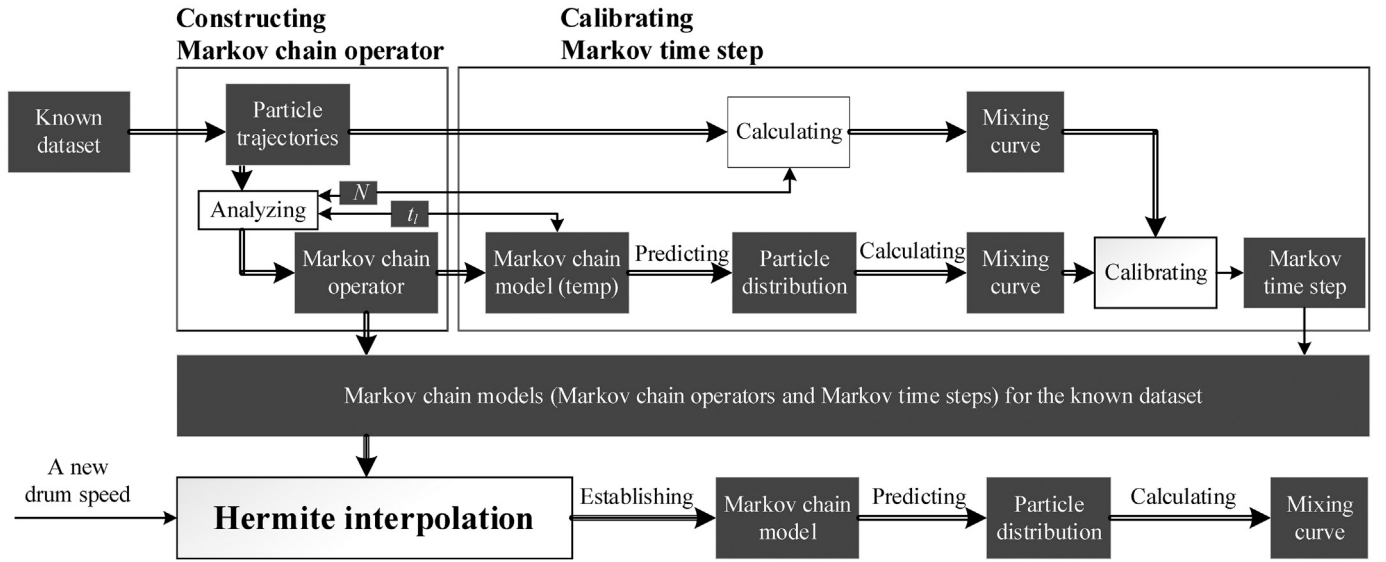


Fig. 2. The schematic diagram of the proposed method.

3.1. Markov chain representation of granular mixing

3.1.1. Markov state and Markov chain operators

The main feature of Markov chain is “Markov property”. Namely, when the current state of the system is known, the future development of the system state is only related to the current state and not affected by its previous states [18,23]. To describe the state of the particle movement, the computational domain Ψ of the drum is divided into N cells (see Fig. 3 where $N = 352$). With the movement of particles, the particle quantities in each cell may change. The *ratio* of the particle quantities N_{cell} in a cell to the particle quantities N_p in domain Ψ , or particle distribution, is usually adopted as a state variable s of the cell (s is ranged

between 0 and 1). For example, the state variable of the cell 1 is calculated as $s^1 = N_{cell\ 1} / N_p$, where $N_{cell\ 1}$ is the total particle quantities in cell 1. The state variables of all the N cells at k -th time step then constitute the Markov state $\mathbf{s}_k = [s_k^1\ s_k^2\ \dots\ s_k^N]^T$ of the granular system. Based on the DEM dataset in Section 2.2, the Markov state of the granular system corresponding to the six specific rotation speeds ($n = 4, 6, 8, 10, 14, 16$ rpm) can be obtained. Image sequence generated by DEM and the corresponding Markov state of the granular bed at $n = 6$ rpm are shown in Table 3 for example.

According to “Markov property”, the state $\mathbf{s}_{k+1}$ is determined by its previous state $\mathbf{s}_k$ mapped through a Markov chain operator $\mathbf{P}$:

$$\mathbf{s}_{k+1} = \mathbf{P}\mathbf{s}_k \quad (1)$$

or

$$\mathbf{s}_{k+1} = \mathbf{P}^{k+1}\mathbf{s}_0 \quad (2)$$

where $\mathbf{s}_0$ is the initial state. It is therefore our main task to determine the Markov chain operator $\mathbf{P}$.

In the present work, we assume that the transfer process is stationary, the transition probability of a particle transferring from cell i to cell j is thus constant at each time step, i.e., $p_{ij, k} = p_{ij}$. Such an assumption was also adopted in [19,20,22,23] and was proven to be valid. Accordingly, the Markov chain operator $\mathbf{P}$ can be given as

$$\mathbf{P} = \begin{bmatrix} p_{11} & p_{12} & \dots & p_{1i} & \dots & p_{1N} \\ p_{21} & p_{22} & \dots & p_{2i} & \dots & p_{2N} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ p_{ij} & p_{2j} & \dots & p_{ij} & \dots & p_{Nj} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ p_{1N} & p_{2N} & \dots & p_{iN} & \dots & p_{NN} \end{bmatrix} \quad (3)$$

According to [22], for cells with stable particle transfer, the Markov chain operator $\mathbf{P}$ has the following properties:

$$p_{ij} \geq 0, \sum_{j=1}^N p_{ij} = 1, \forall i \in \{1, 2, \dots, N\} \quad (4)$$

To determine the value of each element in the Markov chain operator, an indication function $\phi_p(i, k)$ is introduced. If particle p is in the cell i

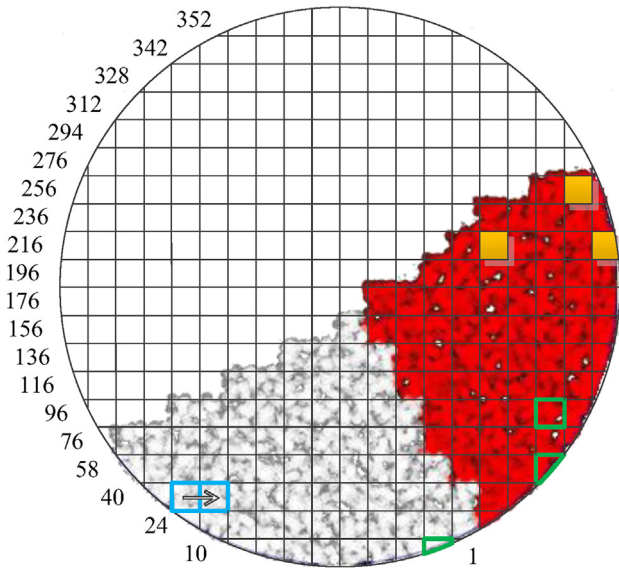
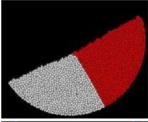
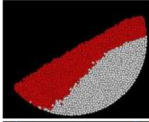
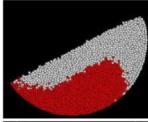
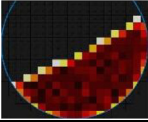
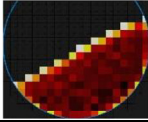
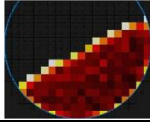
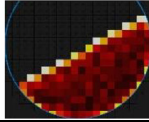
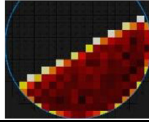


Fig. 3. The computational domain Ψ with N cells ($N = 352$). The number in the figure represents the cell label; the blue rectangles are cell 37# and 38# used in Fig. 5 in Section 3.2; the green rectangles are cell 2#, 42# and 79# used in the example in Section 4.1; the orange squares are cell 197#, 201# and 238# used in the example in Section 4.2.

Table 3Image sequence generated by DEM and the corresponding Markov state of the granular bed at $n = 6$ rpm.

	$t=0s$	$t=2s$	$t=4s$	$t=6s$	$t=10s$	$t=20s$
Mixing image						
Markov state						

*Markov state of the granular bed is composed of state variables of all cells. The value of state variable ranges from 0.17×10^{-3} to 9.08×10^{-3} (the darker color indicates bigger value).

at k -th time step, then $\phi_p(i, k) = 1$; otherwise, $\phi_p(i, k) = 0$. The particle quantity $\varphi(i, k)$ in cell i at k -th time step is

$$\varphi(i, k) = \sum_{p=1}^{N_p} \phi_p(i, k) \quad (5)$$

where N_p is the total particle quantities in domain Ψ . The element p_{ij} in the operator $\mathbf{P}$ can be then calculated as

$$p_{ij} = \frac{1}{N_{LT}} \sum_{k=1}^{N_{LT}} \frac{1}{\varphi(i, k-1)} \sum_{p=1}^{N_p} \phi_p(j, k) \phi_p(i, k-1) \quad (6)$$

where N_{LT} is the number of time steps needed to establish the operator. According to references [22, 28], N_{LT} should be determined so that the quantization error ε between $\mathbf{P}(N_{LT})$ and $\mathbf{P}(N_{LT} + 1)$ is small enough ($\varepsilon < 0.01$ is adopted in the present work)

$$\varepsilon = \frac{\|\mathbf{P}(N_{LT}) - \mathbf{P}(N_{LT} + 1)\|_1}{\|\mathbf{P}(N_{LT})\|_1} \quad (7)$$

This process is called the “learning process” of Markov chain modeling. The learning process begins “when permanent regime conditions prevail in the mixing system” [22]. In the present work, the learning process is started after the particle mixture reaches a steady state, which can be calculated by the method defined in reference [29]. And the time step for learning process, or the learning time step, is set to 0.4 s in the present work, which is equal to the twice time interval in DEM data sampling.

3.1.2. Particle transference

For a granular system at a given rotation speed, if the corresponding Markov chain operators are given, the particle transference at each step can be estimated using Monte-Carlo method to avoid transferring the non-integer quantity of particles. The procedures are:

Step 1: Count the particle quantities in the current cell i , and generate the corresponding quantity of random numbers in $[0, 1]$;

Step 2: Calculate the probability range $[\sum_{l=1}^{j-1} p_{il}, \sum_{l=1}^j p_{il}]$ of the particles transferring from cell i to cell j ;

Step 3: If the random number generated in Step 1 is within the probability range, then transfer the particles from cell i to cell j .

By performing the above procedures, the transference of particles from cell to cell can be completed. Correspondingly, the proportion of red particles in each cell can be calculated.

3.1.3. Description of mixing degree

The degree of particle mixing can be then described on macroscopic level using relative standard deviation (RSD) [22,23]:

$$RSD = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 / \bar{x}} \quad (8)$$

where x_i is the proportion of red particles in cell i , $\bar{x}$ is the proportion of red particles in the computational domain Ψ . In the present work, we set $\bar{x} = 0.5$ because the number of red and white particles is 1:1.

The time evolution of RSD (or RSD curve) is therefore an effective way to describe the dynamic process of granular mixing. A lower

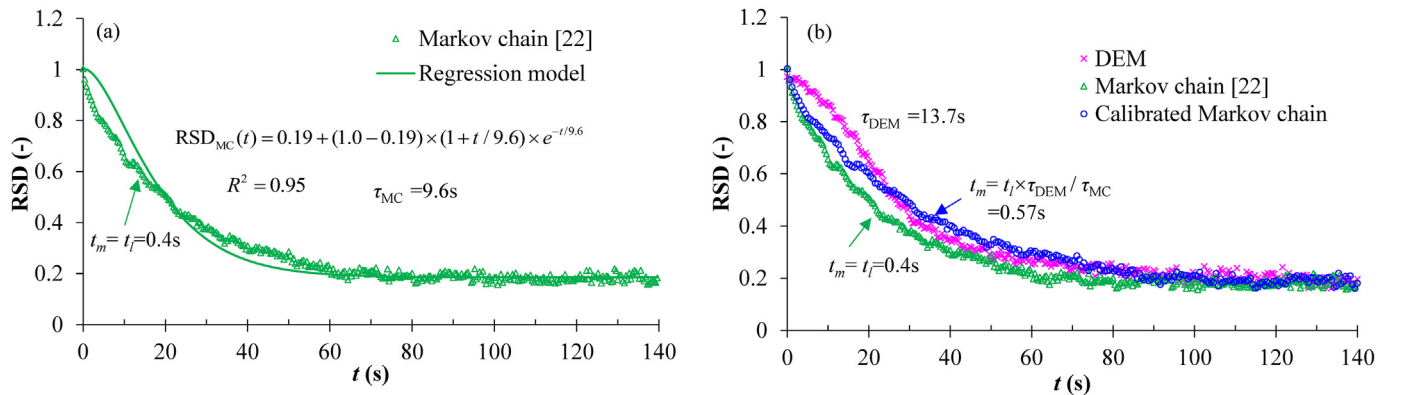


Fig. 4. RSD curves and calibration of Markov time step for the case $n = 6$ rpm, (a) RSD curve obtained by Markov chain model [22] ($t_m = 0.4$ s), (b) RSD curve obtained by DEM, Markov chain model [22] and calibrated Markov chain model.

value of RSD means the red/white particles are better mixed. The RSD curve can be approximated using a second-order regression model [29]:

$$\text{RSD}(t) = \text{RSD}_\infty + (\text{RSD}_0 - \text{RSD}_\infty) \times (1 + t/\tau) \times e^{-t/\tau} \quad (9)$$

where RSD_0 is the initial value, RSD_∞ is the steady state value, and τ is the time constant characterizing the mixing process.

Fig. 4(a) shows for example the RSD curve for the granular mixing process at $n = 6$ rpm. In calculating RSD values by Eq. (8), the proportion x_i of red particles in each cell was determined based on Markov chain operators and the method described in Section 3.1.2. Regression result indicates that the time constant in this case is about 9.6 s.

3.1.4. Calibration of Markov time step

The Markov time step, which is defined as the time between the k -th step and $(k + 1)$ -th step, will affect the result of Markov chain method [22]. In literature [22,23], Markov time step (t_m) was simply set equal to the learning time step (t_l), i.e. $t_m = t_l$, which leads to inaccuracy of predicted results. In order to improve the prediction accuracy, Doucet [22] tried to adjust t_l so that the RSD curve obtained by the Markov chain model is close to that of DEM. However, it is troublesome and time-consuming.

In order to obtain a suitable Markov time step, we calibrate the time step by:

$$t_m = t_l \times \tau_{\text{DEM}}/\tau_{\text{MC}} \quad (10)$$

where τ_{MC} is the time constant obtained by a temporary Markov chain model whose Markov time step is set equal to t_l . τ_{DEM} is the time constant obtained from RSD curves generated by DEM.

Fig. 4(b) shows for example the RSD curves obtained by Markov chain method [22] and DEM for the case $n = 6$ rpm. When the Markov time step is set equal to the learning time step, i.e., $t_m = t_l = 0.4$ s, the predicted RSD curve shows much deviation from that of DEM. With the adjusted Markov time step calculated by Eq. (10) where $t_m = 0.57$ s, the deviation becomes much smaller. The calibrated Markov time step for other five cases ($n = 4, 8, 10, 14, 16$ rpm) in the known dataset is listed in Table 4.

3.2. Hermite interpolation

By use of the method described above, the Markov chain operators and Markov time step for the six specific drum speeds (4, 6, 8, 10, 14, 16 rpm) can be obtained. However, they cannot be directly adopted for predicting the granular mixing process at new rotation speeds. To solve this problem, we use interpolation method to construct new Markov chain operators and Markov time step for the new drum speed.

Hermite interpolation method is adopted in the present work. Denote $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_k, \mathbf{P}_{k+1}, \dots, \mathbf{P}_m$ as the already established Markov chain operators for the specific rotation speed $n_1, n_2, \dots, n_k, n_{k+1}, \dots, n_m$, respectively ($m = 6$ in the present work). Based on these operators, the new Markov chain operator $\mathbf{P}^*$ for a new drum speed n ($n_k < n < n_{k+1}$) can be approximated using piece-wise cubic Hermite interpolation (PCHI) [30]:

Table 4
Calibrated Markov time steps for the known dataset.

No.	Rotation speed (rpm)	Time step of the learning process t_l (s)	Markov time step t_m (s)
1	$n_1 = 4$	0.4	0.84
2	$n_2 = 6$	0.4	0.57
3	$n_3 = 8$	0.4	0.50
4	$n_4 = 10$	0.4	0.49
5	$n_5 = 14$	0.4	0.48
6	$n_6 = 16$	0.4	0.44

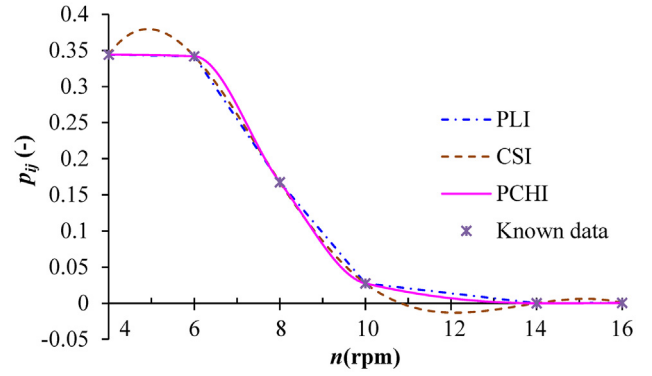


Fig. 5. Interpolation results of transition probability p_{ij} (from cell 38# to cell 37#) at different rotation speeds by piece-wise linear interpolation (PLI), cubic spline interpolation (CSI) and piece-wise cubic Hermite interpolation (PCHI).

$$\begin{aligned} \mathbf{P}^* = & \left(1 + 2 \frac{n - n_k}{n_{k+1} - n_k}\right) \left(\frac{n - n_{k+1}}{n_k - n_{k+1}}\right)^2 \mathbf{P}_k \\ & + \left(1 + 2 \frac{n - n_{k+1}}{n_k - n_{k+1}}\right) \left(\frac{n - n_k}{n_{k+1} - n_k}\right)^2 \mathbf{P}_{k+1} \\ & + (n - n_k) \left(\frac{n - n_{k+1}}{n_k - n_{k+1}}\right)^2 \mathbf{d}_k + (n - n_{k+1}) \left(\frac{n - n_k}{n_{k+1} - n_k}\right)^2 \mathbf{d}_{k+1} \end{aligned} \quad (11)$$

where $\mathbf{P}_k$ and $\mathbf{P}_{k+1}$ is the Markov chain operator corresponding to drum speed n_k and n_{k+1} . $\mathbf{d}_k$ and $\mathbf{d}_{k+1}$ is the derivative matrix of $\mathbf{P}_k$ and $\mathbf{P}_{k+1}$, which can be calculated by the algorithm described in reference [30].

By use of Eq. (11), the matrix $\mathbf{P}^*$ can be obtained and thus the transition probability p_{ij} of particles at a given new drum speed n can be determined. Fig. 5 shows for example the interpolation result of transition probability p_{ij} (from cell 38# to cell 37#) at different drum speeds. For comparison, results by other interpolation methods (piece-wise linear interpolation (PLI) and cubic spline interpolation (CSI)) are also illustrated in this figure. It is shown that the curve by PCHI method is smoother than that of PLI, while CSI may lead to wrong result (negative value of transition probability).

It should be pointed out that, the prediction accuracy of the proposed method will be deteriorated in the case of *extrapolation* where the new drum speed is far beyond the scope of the drum speeds of the known dataset. So, we recommend *interpolation*, that is, the new rotation speed is within the scope of the rotation speeds of the known dataset.

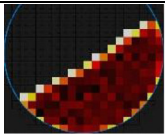
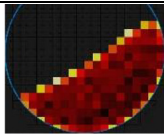
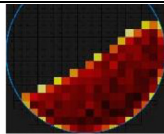
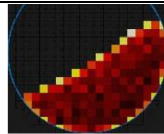
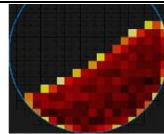
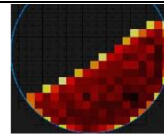
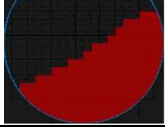
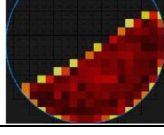
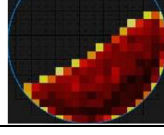
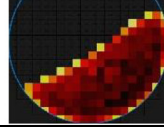
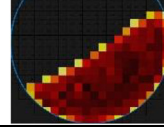
Although the Markov chain operator $\mathbf{P}^*$ can be approximated using Eq. (11), it may not satisfy the condition $\sum_{j=1}^N p_{ij} = 1$ listed in Eq. (4). Therefore, two post-processing procedures are taken in the present work:

- (1) determine whether each cell in the computational domain Ψ is activated or not. Given cell i , all elements of the i -th column in $\mathbf{P}^*$ are summed; if the sum is greater than a threshold (0.5 in this work), cell i is considered as activated, otherwise, it is inactivated.
- (2) the corresponding column of $\mathbf{P}^*$ is normalized for the activated cell and set to 0 for inactivated cell. By doing so, the $\mathbf{P}^*$ can satisfy the Markov chain properties required by Eq. (4).

Similarly, the Markov time step needed for a new drum speed n can be also obtained by Hermite interpolation, based on the data listed in Table 4.

By use of the Hermite interpolation-based Markov chain method (HI-MCM) described in Section 3, it is now possible for us to predict

Table 5
Markov states of the granular bed at fixed drum speed ($n = 12.34$ rpm).

Method	$t=0s$	$t=2s$	$t=6s$	$t=10s$	$t=20s$	$t=40s$
DEM						
Proposed method						

* The state variable (particle distribution) is ranged between 0.17×10^{-3} to 9.08×10^{-3} (darker color indicates bigger value).

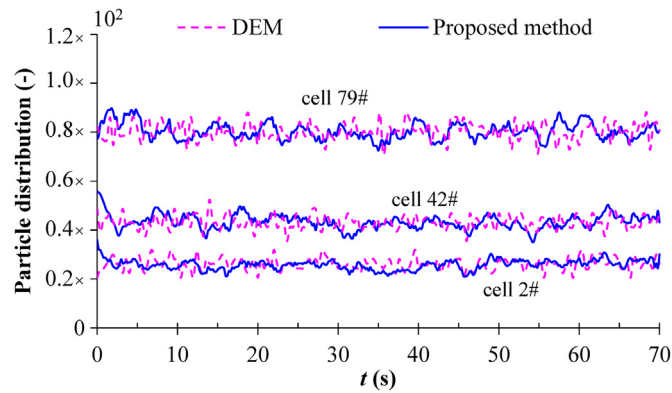


Fig. 6. The time evolutions of particle distribution in cells 2#, 42#, 79# obtained by DEM and the proposed method ($n = 12.34$ rpm).

granular mixing process at new rotation speeds, without performing additional time-consuming DEM simulations. In the following sections, the predicted results by HI-MCM will be compared with results by DEM and other Markov chain methods.

4. Results

4.1. Prediction of granular mixing at fixed drum speed

Granular mixing at fixed drum speed means that the speed is kept constant after the rotation begins. Although the proposed method (HI-MCM) is established based on data from six DEM simulations (i.e., the known dataset No.1–No.6 in Table 2) where the drum speeds are respectively fixed at $n = 4, 6, 8, 10, 14, 16$ rpm, it is expected to be capable of predicting mixing process at new rotation speeds. In the following texts, we will use the new rotation speed $n = 12.34$ rpm as example.

In Table 5, the Markov states of the granular bed predicted by the proposed method for the new speed $n = 12.34$ rpm are visualized and compared with those obtained from DEM simulation. The Markov time step for $n = 12.34$ rpm is calculated as $t_m = 0.49$ s using Hermite interpolation. In DEM method, the particles are generated by the sedimentation method, which makes the state variable (particle distribution) of each cell different at $t = 0$; in the proposed method, the state variable of each cell at initial time is set to be the same ($s_i = 6.76 \times 10^{-3}$). That is why the visualized Markov states of DEM method and the proposed method are different at $t = 0$. However, with the rotation of the drum, the predicted Markov states by these two methods seem similar. For quantitative comparison, the time evolution of the particle distribution (or state variable) of some cells (cell 2#, 42# and 79# in Fig. 3) are illustrated in Fig. 6. It is shown that the particle distribution in each cell

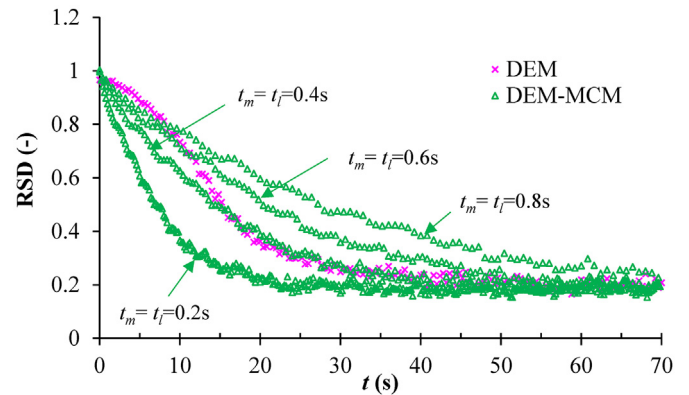


Fig. 7. RSD curves obtained by other Markov chain method (DEM-MCM) with different learning time step t_l ($n = 12.34$ rpm).

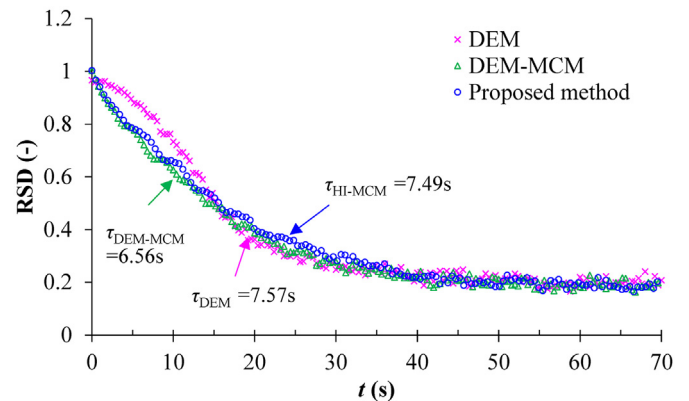


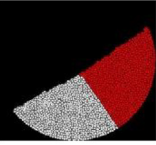
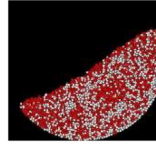
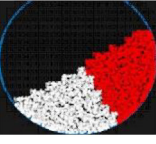
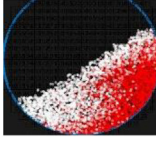
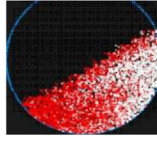
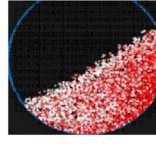
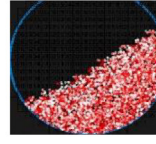
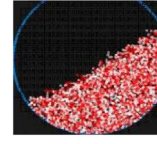
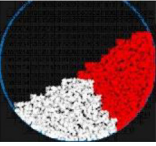
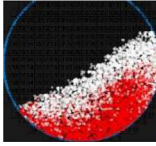
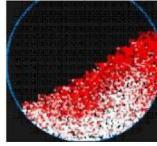
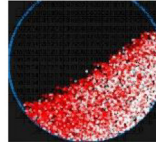
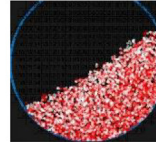
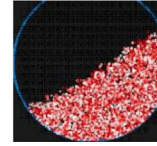
Fig. 8. Mixing process predicted by DEM, DEM-MCM and the proposed HI-MCM method ($n = 12.34$ rpm).

fluctuates around a constant and there is little difference between the values predicted by proposed method and DEM.

With known Markov state of the granular bed, the RSD mixing curve can be obtained using the proposed method, which will be then compared with results of the other Markov chain method (DEM-MCM) and DEM method.

The DEM-MCM method needs to perform DEM simulation to predict the mixing process at a new drum speed. The main steps are as follows [22]: a) performing DEM simulation for the new drum speed $n = 12.34$ rpm. The DEM simulation was performed for physical time 22.4 s to obtain the particle trajectories at the steady state of mixing

Table 6Comparison of predicted image sequences obtained by DEM, DEM-MCM, and proposed method ($n = 12.34$ rpm).

Method	$t = 0s$	$t = 6s$	$t = 10s$	$t = 20s$	$t = 40s$	$t = 70s$
DEM						
DEM-MCM						
Proposed method						

process; b) establishing Markov chain model. The Markov chain operator is constructed with the learning time step t_l by the “learning process”; c) implementing the Markov chain model to predict the particle distribution. The RSD curve can be then obtained. In DEM-MCM, t_m is set equal to t_l . In order to get higher prediction accuracy, t_l needs to be adjusted multiple times, as shown in Fig. 7. For $t_m = t_l = 0.4$ s, the RSD curve predicted by DEM-MCM is closest to that by DEM.

In contrast, our proposed method does not need to perform new round of DEM simulations when predicting granular mixing at new rotation speed. The predicted RSD curve for $n = 12.34$ rpm is shown in Fig. 8 and compared with that of DEM method and DEM-MCM method. The RSD curves start from about 1 where the red and white particles are separated; as the drum rotates, the particles begin to mix, leading to rapid drop of RSD until a steady state (about 0.2) is reached. The time constant of the mixing process predicted by the proposed method ($\tau_{HI-MCM} = 7.49$ s) is in good agreement with DEM result ($\tau_{DEM} = 7.57$ s), while the DEM-MCM result ($\tau_{DEM-MCM} = 6.56$ s) has a relative error of 13%. It is because the proposed method has a step to calibrate the Markov time step, which makes the predicted mixing process closer to that of DEM.

By use of the proposed method, the image sequence of the mixing process can be also obtained with following procedures: (i) generate particles with a specified number in each activation cell (the white particles are on the left and the red particles are on the right, as shown in

the image at $t = 0$ s in Table 6) and number them; (ii) calculate the position of each particle (labeled red or white) at different time, using the steps described in Section 3.1.2; (iii) the calculated positions and color attributes of all particles at each time step are then visualized using MATLAB *plot* and *fill* function. In this way, we can obtain the image sequence of the mixing process shown in Table 6. It should be pointed out here that the specific position of the transferred particles in the cells is random in the present work. From Table 6, we can see that the mixing pattern predicted by Markov chain methods (DEM-MCM and the proposed method) is more developed for the first 10 s than that of DEM. It is because Markov chain methods provide the particle quantity in each cell, and the particle position in each cell are randomly arranged. Doucet et al. [22] found the mixing pattern can be closer to that of DEM by increasing the cell number N .

Although the accuracy of the DEM-MCM method and our proposed method is similar in predicting mixing patterns, our proposed method is more advantageous in computation speed, as shown in Table 7. For the new drum speed $n = 12.34$ rpm, the proposed method takes only 87.9 s to predict the first 70 s of the mixing process, whereas DEM-MCM needs 181.5 min and DEM costs about 553.4 min. It is because the proposed method needs no more time-consuming DEM simulation, and the time for constructing the Markov chain model by Hermite interpolation is also shorter than by the “learning process”.

4.2. Prediction of granular mixing with varying drum speeds

Granular mixing with varying drum speeds means that the speed is changed after the mixing has already begun. The proposed method is also effective in this case. Suppose that the drum has been operated for k_1 time steps with rotation speed n_1 from the initial state $\mathbf{s}_0$ and then the rotation speed is adjusted to n_2 for k_2 time steps. The Markov state $\mathbf{s}_k$ at k -th time step ($k = k_1 + k_2$) can be obtained by:

$$\mathbf{s}_k = \mathbf{P}_{n_2}^{k_2} \mathbf{P}_{n_1}^{k_1} \mathbf{s}_0 \quad (12)$$

where, $\mathbf{P}_{n_1}$ and $\mathbf{P}_{n_2}$ are the Markov chain operators for the drum speeds n_1 and n_2 respectively. $\mathbf{P}_{n_1}$ and $\mathbf{P}_{n_2}$ can be quickly obtained by Hermite interpolation described in Section 3.2.

Fig. 9(a) shows for example the periodic change of rotation speed from 15 rpm to 5 rpm, with a time interval of 20s. The drum runs for 3 cycles. The Markov chain operators ($\mathbf{P}_{15}$ and $\mathbf{P}_5$) and Markov time

Table 7Computational expense for predicting the mixing process of the first 70 s ($n = 12.34$ rpm).

Method	Procedures	Computing time
DEM	DEM simulation.	553.4 mins (in total)
DEM-MCM	a) Performing DEM simulation	169.0 mins
	b) Establishing Markov chain model	11.5 mins
	c) Implementing the Markov chain model	60.7 s
Proposed method		181.5 mins (in total)
	a) Establishing Markov chain model	37.2 s
	b) Implementing the Markov chain model	50.7 s
		87.9 s (in total)

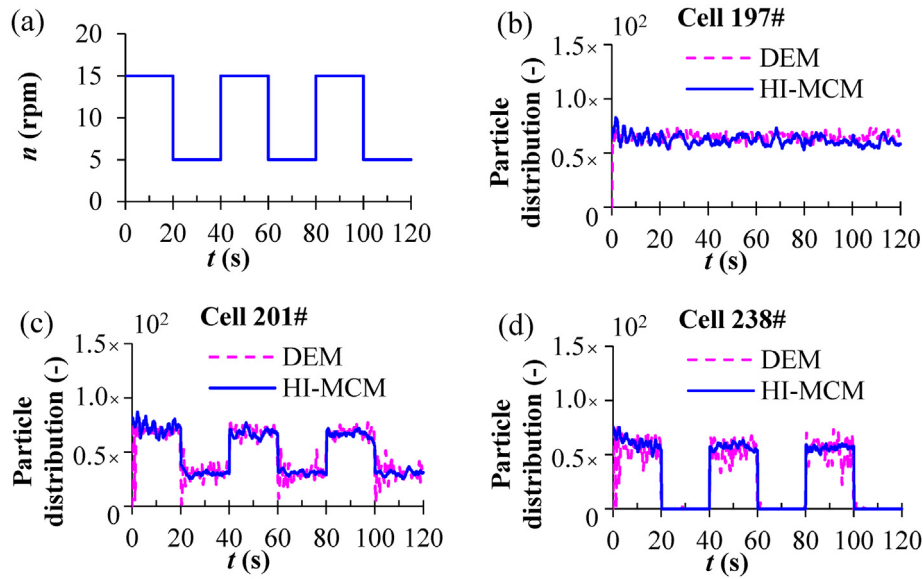


Fig. 9. (a) The setting of drum speed; (b-d) the particle distributions under varying drum speeds in cells 197#, 201#, and 238#, respectively.

step ($t_{m,15}=0.47$ s and $t_{m,5}=0.67$ s) corresponding to 15 rpm and 5 rpm can be quickly obtained by Hermite interpolation. The Markov states of the granular bed at varying drum speeds are visualized in Table 8. The result is in good agreement with that of DEM method.

The time evolution of particle distributions at varying drum speeds is shown in Fig. 9, taking cells in different bed regions as example (cell 197#, 201# and 238# in Fig. 3). It can be seen the particle distributions obtained by the proposed method (HI-MCM) still maintain consistency with those obtained by DEM. For cells in the passive region of the bed (cell 197#), the particle distribution fluctuates around a constant value (see Fig. 9(b)) because the particles rotate as a rigid body with the same angular velocity of the drum wall. For cells in the active region of the bed (cell 201#, cell 238#), the particle distribution changes periodically with drum speed due to the avalanching of particles, as shown in Fig. 9(c) and (d).

The corresponding RSD curve at varying rotation speed is shown in Fig. 10, and compared with that obtained by DEM and the other Markov chain method (Tjakra's method [23]). Tjakra's method is an extension of the DEM-MCM method, which also needs to perform DEM simulations for the new varying drum speeds before the method can be used. In the case of periodic change of rotation speed shown in Fig. 9(a), Tjakra's

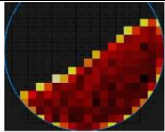
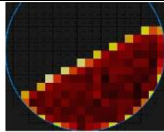
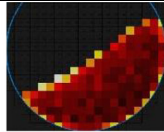
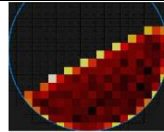
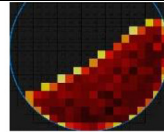
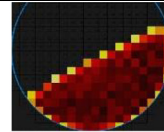
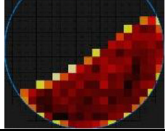
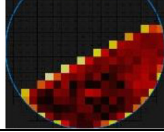
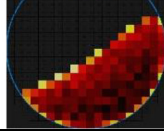
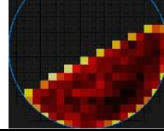
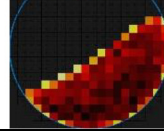
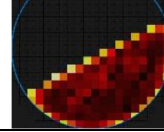
method needs to perform two DEM simulations for physical time 20s ($n_1=15$ rpm) and 22.4 s ($n_2=5$ rpm), respectively; the two corresponding Markov chain operators (P_{15} and P_5) and Markov time step ($t_{m,15}=0.4$ s and $t_{m,5}=0.6$ s) can be then obtained; finally, the particle distribution and thus RSD curve under varying drum speed can be calculated by Eq. (12).

It can be seen from Fig. 10 that RSD curve obtained by proposed method is consistent with the curve obtained by DEM under varying drum speed. And the mixing time constant obtained by proposed method ($\tau_{HI-MCM} = 6.86$ s) is closer to that of DEM ($\tau_{DEM} = 7.0$ s) than that obtained by Tjakra's method ($\tau_{Tjakra} = 6.66$ s). Moreover, the proposed method outperforms Tjakra's approach in computing efficiency, in that its computational expense is less than 1% of that by Tjakra's method, as shown in Table 9. It has potential usages in online prediction of granular mixing in rotary drums.

In summary, compared with current Markov-chain based methods (the DEM-MCM method and Tjakra's method), the advantages of our proposed HI-MCM method are that: (i) it can predict the granular mixing at fixed drum speed as well as granular mixing with varying drum speeds. (ii) in predicting granular mixing at a new rotation speed, both DEM-MCM method and Tjakra's method need the DEM

Table 8

Predicted Markov state of the granular bed by DEM and the proposed method for varying drum speeds.

Method	$t=20s$	$t=40s$	$t=60s$	$t=80s$	$t=100s$	$t=120s$
DEM						
Proposed method						

* The state variable (particle distribution) is ranged between 0.17×10^{-3} to 9.08×10^{-3} (darker color indicates bigger value).

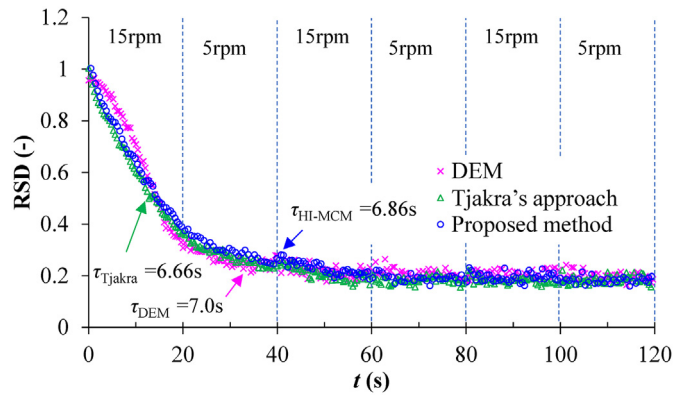


Fig. 10. RSD curves predicted by DEM, Tiakra's method and the proposed HI-MCM method under varying rotation speeds.

Table 9

Computational expense for predicting the granular mixing process of 120 s at the new varying drum speeds.

Method	Procedures	Computing time
DEM	DEM simulation	736.7 mins (in total)
Tjakra's approach	a) Performing DEM simulation b) Establishing Markov chain model c) Implementing the Markov chain model	370.6 mins 29.7 mins 96.1 s
		401.9 mins (in total)
Proposed method	a) Establishing Markov chain model b) Implementing the Markov chain model	77.6 s 72.2 s
		149.8 s (in total)

simulation data at this new rotation speed, that is, a new round of time-consuming DEM simulation must be performed before these two methods can be used. In contrast, this is not necessary in our proposed method, therefore, it costs much less computing time.

5. The influence of Markov chain parameters on the proposed method

In this section, the influence of two important Markov chain parameters (the cell number N and the learning time step t_l) on the predicted results will be discussed, taking the new drum speed $n = 12.34$ rpm as an example.

5.1. Influence of the cell number

Fig. 11 shows the RSD mixing curves predicted by the proposed method using different cell numbers ($N = 96, 352, 1324$). As N varies from 96 to 1324, the relative error between the time constant predicted by the proposed method (τ_{HI-MCM}) and that by DEM (τ_{DEM}) is less than 5%. And, RSD curve obtained by the proposed method is close to the curve obtained by DEM with increasing cell number. However, the Markov time step t_m becomes smaller with increasing cell number, which means more computing time.

The image sequences of mixing process predicted by the proposed method using different cell numbers are compared in Table 10. As the cell number is increased from 96 to 1324, the image sequence predicted by our method becomes closer to DEM result. Nevertheless, the computing time needed is also dramatically increased from 21.9 s to 648.2 s. The cell number $N = 352$ is selected in the present work, which is a choice of balance between prediction accuracy and computing efficiency. It should be noted that, the number of particles in the activated cell should be more than 10 so that the influence of noise can be reduced [31].

5.2. Influence of the learning time step

Learning time step t_l is another model parameter in our proposed HI-MCM method. Its influence on the prediction results is shown in Table 11. The cell number is fixed at $N = 352$, while five different values of learning time step are adopted in the calculations ($t_l = 0.2$ s, 0.4 s, 0.8 s, 1.6 s, 3.2 s). It is found the learning time step has no obvious influence on the Markov time step, the time constant of the mixing process, and the regression accuracy of the RSD curve. This is because an additional step for calibrating the Markov time step t_m is introduced in our

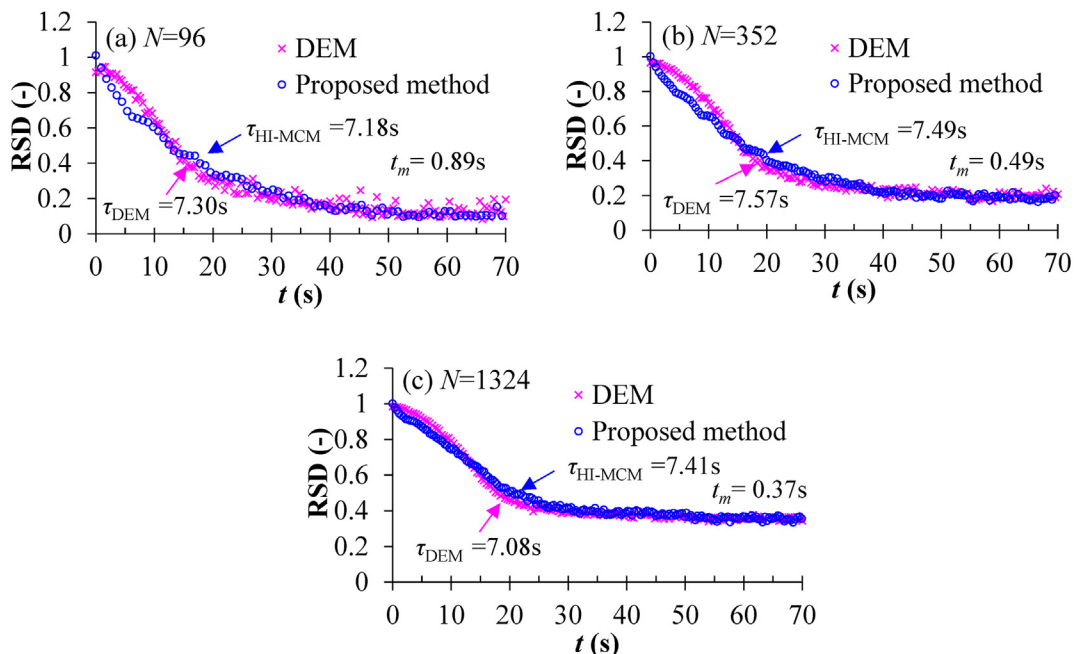


Fig. 11. RSD curves obtained by DEM and the proposed HI-MCM method under different cell numbers N ($n = 12.34$ rpm), (a) $N = 96$, (b) $N = 352$, (c) $N = 1324$.

Table 10Image sequences obtained by the proposed method using different cell numbers N ($n = 12.34$ rpm).

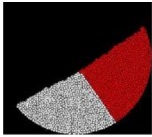
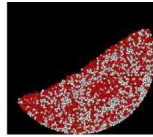
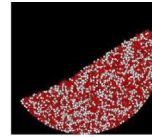
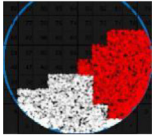
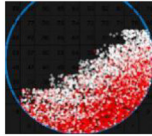
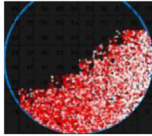
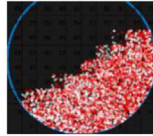
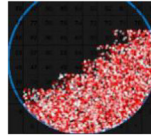
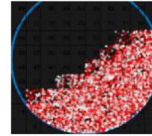
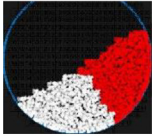
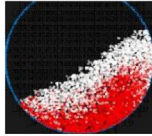
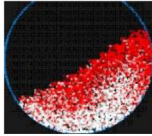
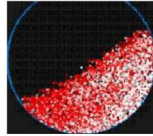
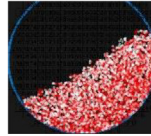
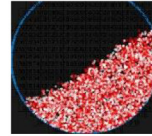
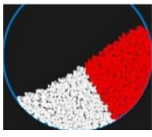
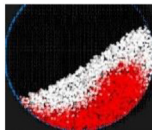
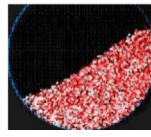
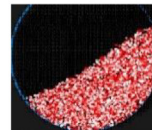
Method	$t = 0s$	$t = 6s$	$t = 10s$	$t = 20s$	$t = 40s$	$t = 70s$	Computing time
DEM							553.4 min
Proposed method ($N=96$)							21.9 s
Proposed method ($N=352$)							87.9 s
Proposed method ($N=1324$)							648.2 s

Table 11Quantitative comparison of mixing process predicted by the proposed HI-MCM method under different learning time steps t_l for 70s ($N = 352$, $n = 12.34$ rpm).

	HI-MCM	HI-MCM	HI-MCM	HI-MCM	HI-MCM	DEM
Learning time step t_l (s)	0.2	0.4	0.8	1.6	3.2	–
Markov time step t_m (s)	0.48	0.49	0.48	0.46	0.47	–
Time constant τ (s)	7.75 ± 0.13	7.49 ± 0.13	7.73 ± 0.13	7.85 ± 0.14	7.87 ± 0.13	7.57 ± 0.08
RSD regression accuracy R^2 (–)	0.98	0.97	0.97	0.97	0.98	0.98

HI-MCM method, so that the predicted result is robust to the change of learning time step.

6. Conclusions

DEM is a powerful but computationally intensive method in simulating granular mixing process in rotary drums. When the operating condition (such as rotation speed) of the drum is changed, a new round of time-consuming DEM simulation must be performed. For faster prediction of granular mixing at new operating parameters, we proposed a novel method (HI-MCM). HI-MCM utilizes the known dataset (data generated by only several DEM simulations) to establish Markov chain models, with which the mixing process at new drum speeds can be then quickly predicted by Hermite interpolation. It is not necessary to perform additional DEM simulations. Compared with current Markov chain methods the advantages of the proposed method include:

- (1) HI-MCM can predict the granular mixing at fixed drum speed as well as at varying drum speeds, with no need to perform additional DEM simulations. However, HI-MCM belongs to data-driven modelling method using interpolation. In order to guarantee prediction accuracy, we recommend the new rotation speeds to be predicted should not be beyond the scope of rotation speeds in the known dataset.

- (2) The mixing process (in terms of mixing pattern and RSD mixing curves) predicted by HI-MCM is in good agreement with DEM result, while the computing time is dramatically reduced (seconds vs. hours). It has potential usages in online prediction of granular mixing in rotary drums.
- (3) HI-MCM outperforms other Markov chain methods in computing speed. Comparison demonstrates that the computing time needed by HI-MCM is less than 1% of that needed by the DEM-MCM method [22] and Tjakra's method [23].
- (4) Different to other Markov chain methods, HI-MCM method introduces an additional step to calibrate the Markov time step. In this way, HI-MCM has good robustness to the change of learning time step. However, like other Markov chain models, cell number N is a model parameter affecting the prediction results of the proposed method. Higher values of N lead to improved accuracy but more computing time, a balance between them should be made.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Algorithm

Algorithm of the proposed HI-MCM method

```
// Establish the Markov chain models for the known dataset
1: for each drum speed  $n_m$  in the known dataset do
2: Construct the Markov chain operator  $P_m$  by Eq. (6) and Eq. (7)
3: Calibrate the Markov time step  $t_m$  by Eq. (10)
4: end for
// Establish the new Markov chain model for a new rotation speed
5: Construct the new Markov chain operator  $P^*$  and the corresponding Markov time step  $t^*$  by Hermite interpolation
6: Predict the particle distribution  $S_k$  using the interpolated model by Step 5, then calculate the RSD curve by Eq. (8)
```

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